PADHAKU
Mathematics · Class 10 Mathematics



CHAPTER 12

Surface Areas and Volumes

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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Exercise 12.1

1 **2 cubes each of volume 64 cm³ are joined end to end. Find the surface area of the resulting cuboid.**

✔ ANSWER

Volume of each cube = 64 cm³, so side = $\sqrt[3]{64}$ = 4 cm.

Joined end to end, the cuboid is 8 cm × 4 cm × 4 cm (l = 8, b = 4, h = 4).

Surface area = 2(lb + bh + hl) = 2(8×4 + 4×4 + 4×8) = 2(32 + 16 + 32) = 2 × 80

Surface area of the cuboid = 160 cm²

2 **A vessel is in the form of a hollow hemisphere mounted by a hollow cylinder. The diameter of the hemisphere is 14 cm and the total height of the vessel is 13 cm. Find the inner surface area of the vessel.**

✔ ANSWER

Radius of hemisphere = radius of cylinder, r = 14/2 = 7 cm.

Height of cylinder h = 13 − 7 = 6 cm.

Inner surface area = CSA of hemisphere + CSA of cylinder = $2\pi r^2 + 2\pi rh = 2\pi r(r + h)$

= 2 × 22/7 × 7 × (7 + 6) = 44 × 13

Inner surface area = 572 cm²

3 **A toy is in the form of a cone of radius 3.5 cm mounted on a hemisphere of same radius. The total height of the toy is 15.5 cm. Find the total surface area of the toy.**

✔ ANSWER

r = 3.5 cm. Height of cone h = 15.5 − 3.5 = 12 cm.

Slant height l = $\sqrt{(r^2 + h^2)} = \sqrt{(12.25 + 144)} = \sqrt{156.25} = 12.5$ cm.

CSA of cone = $\pi rl = 22/7 \times 3.5 \times 12.5 = 137.5$ cm²

CSA of hemisphere = $2\pi r^2 = 2 \times 22/7 \times 3.5 \times 3.5 = 77$ cm²

Total surface area = 137.5 + 77

TSA of the toy = 214.5 cm²

4 **A cubical block of side 7 cm is surmounted by a hemisphere. What is the greatest diameter the hemisphere can have? Find the surface area of the solid.**

✔ ANSWER

The hemisphere sits on a face of the cube, so its diameter can be at most the side of the cube.

Greatest diameter = 7 cm, so r = 3.5 cm.

Surface area = TSA of cube − base area of hemisphere + CSA of hemisphere

= $6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$

= 6 × 7 × 7 + 22/7 × 3.5 × 3.5 = 294 + 38.5

Surface area of the solid = 332.5 cm²

5 **A hemispherical depression is cut out from one face of a cubical wooden block such that the diameter l of the hemisphere is equal to the edge of the cube. Determine the surface area of the remaining solid.**

✔ ANSWER

Edge of cube = l, so radius of hemisphere r = l/2.

Surface area = TSA of cube − area of circle removed + CSA of hemispherical depression

= $6l^2 - \pi r^2 + 2\pi r^2 = 6l^2 + \pi r^2$

= $6l^2 + \pi(l/2)^2 = 6l^2 + \pi l^2/4$

Surface area = $(l^2/4)(\pi + 24)$ sq. units

6 **A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. The length of the entire capsule is 14 mm and the diameter of the capsule is 5 mm. Find its surface area.**

✔ ANSWER

Diameter = 5 mm, so r = 2.5 mm.

Length of cylindrical part h = 14 − 2.5 − 2.5 = 9 mm.

Surface area = CSA of cylinder + CSA of two hemispheres = $2\pi rh + 2 \times 2\pi r^2 = 2\pi r(h + 2r)$

= 2 × 22/7 × 2.5 × (9 + 5) = 2 × 22/7 × 2.5 × 14 = 220

Surface area of the capsule = 220 mm²

7 **A tent is in the shape of a cylinder surmounted by a conical top. If the height and diameter of the cylindrical part are 2.1 m and 4 m respectively, and the slant height of the top is 2.8 m, find the area of the canvas used for making the tent. Also, find the cost of the canvas of the tent at the rate of ₹ 500 per m². (The base of the tent is not covered with canvas.)**

✔ ANSWER

r = 4/2 = 2 m, h = 2.1 m, slant height of cone l = 2.8 m.

Area of canvas = CSA of cylinder + CSA of cone = 2πrh + πrl = πr(2h + l)

= 22/7 × 2 × (4.2 + 2.8) = 22/7 × 2 × 7 = 44

Area of canvas = 44 m²

Cost = 44 × ₹ 500

Cost of canvas = ₹ 22,000

8 **From a solid cylinder whose height is 2.4 cm and diameter 1.4 cm, a conical cavity of the same height and same diameter is hollowed out. Find the total surface area of the remaining solid to the nearest cm².**

✔ ANSWER

r = 1.4/2 = 0.7 cm, h = 2.4 cm.

Slant height of cone $l = \sqrt{(r^2 + h^2)} = \sqrt{(0.49 + 5.76)} = \sqrt{6.25} = 2.5$ cm.

TSA of remaining solid = CSA of cylinder + area of base circle + CSA of conical cavity

$= 2\pi rh + \pi r^2 + \pi rl = \pi r(2h + r + l)$

$= 22/7 \times 0.7 \times (4.8 + 0.7 + 2.5) = 2.2 \times 8 = 17.6$ cm²

TSA ≈ 18 cm² (to the nearest cm²)

9 **A wooden article was made by scooping out a hemisphere from each end of a solid cylinder. If the height of the cylinder is 10 cm, and its base is of radius 3.5 cm, find the total surface area of the article.**

✔ ANSWER

r = 3.5 cm, h = 10 cm.

TSA = CSA of cylinder + CSA of two hemispheres $= 2\pi rh + 2 \times 2\pi r^2 = 2\pi r(h + 2r)$

$= 2 \times 22/7 \times 3.5 \times (10 + 7) = 22 \times 17$

Total surface area of the article = 374 cm²

Exercise 12.2

1 **A solid is in the shape of a cone standing on a hemisphere with both their radii being equal to 1 cm and the height of the cone is equal to its radius. Find the volume of the solid in terms of π.**

✔ ANSWER

r = 1 cm, height of cone h = 1 cm.

Volume = volume of cone + volume of hemisphere = $\frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$

= $\frac{1}{3}\pi \times 1 \times 1 + \frac{2}{3}\pi \times 1 = \frac{1}{3}\pi + \frac{2}{3}\pi$

Volume of the solid = π cm^3

2 **Rachel was asked to make a model shaped like a cylinder with two cones attached at its two ends using a thin aluminium sheet. The diameter of the model is 3 cm and its length is 12 cm. If each cone has a height of 2 cm, find the volume of air contained in the model. (Assume the outer and inner dimensions to be nearly the same.)**

✔ ANSWER

r = 3/2 = 1.5 cm, height of each cone h_1 = 2 cm.

Height of cylinder h_2 = 12 − 2 − 2 = 8 cm.

Volume of air = volume of cylinder + 2 × volume of cone = $\pi r^2 h_2 + 2 \times \frac{1}{3}\pi r^2 h_1$

= $\pi r^2 (h_2 + \frac{2}{3}h_1)$ = 22/7 × 1.5 × 1.5 × (8 + 4/3)

= 22/7 × 2.25 × 28/3 = 22 × 2.25 × 4/3 = 66

Volume of air in the model = 66 cm³

3 **A gulab jamun contains sugar syrup up to about 30% of its volume. Find approximately how much syrup would be found in 45 gulab jamuns, each shaped like a cylinder with two hemispherical ends with length 5 cm and diameter 2.8 cm.**

✔ ANSWER

r = 2.8/2 = 1.4 cm. Height of cylindrical part h = 5 − 1.4 − 1.4 = 2.2 cm.

Volume of one gulab jamun = $\pi r^2 h + 2 \times \frac{2}{3}\pi r^3$ = $\pi r^2(h + 4r/3)$

= $22/7 \times 1.4 \times 1.4 \times (2.2 + 5.6/3) = 6.16 \times (2.2 + 1.8667) = 6.16 \times 4.0667 \approx 25.05$ cm³

Volume of 45 gulab jamuns ≈ 45 × 25.05 = 1127.28 cm³

Syrup = 30% of 1127.28 ≈ 338.18 cm³

Syrup in 45 gulab jamuns ≈ 338 cm³

4 **A pen stand made of wood is in the shape of a cuboid with four conical depressions to hold pens. The dimensions of the cuboid are 15 cm by 10 cm by 3.5 cm. The radius of each of the depressions is 0.5 cm and the depth is 1.4 cm. Find the volume of wood in the entire stand.**

✔ ANSWER

Volume of cuboid = 15 × 10 × 3.5 = 525 cm³

Volume of one conical depression = $\frac{1}{3}\pi r^2 h = \frac{1}{3} \times 22/7 \times 0.5 \times 0.5 \times 1.4 = 0.3667$ cm³ (= 11/30 cm³)

Volume of 4 depressions = 4 × 11/30 = 1.4667 cm³

Volume of wood = 525 − 1.4667 = 523.5333...

Volume of wood in the stand ≈ 523.53 cm³

5 **A vessel is in the form of an inverted cone. Its height is 8 cm and the radius of its top, which is open, is 5 cm. It is filled with water up to the brim. When lead shots, each of which is a sphere of radius 0.5 cm, are dropped into the vessel, one-fourth of the water flows out. Find the number of lead shots dropped in the vessel.**

✔ ANSWER

Volume of cone = $\frac{1}{3}\pi r^2 h = \frac{1}{3} \times \pi \times 5 \times 5 \times 8 = 200\pi/3$ cm³

Water that flows out = ¼ × 200π/3 = 50π/3 cm³

Volume of one lead shot = 4/3 πr³ = 4/3 × π × (0.5)³ = 4/3 × π × 1/8 = π/6 cm³

Number of shots = (50π/3) ÷ (π/6) = 50/3 × 6 = 100

Number of lead shots = 100

6 **A solid iron pole consists of a cylinder of height 220 cm and base diameter 24 cm, which is surmounted by another cylinder of**

height 60 cm and radius 8 cm. Find the mass of the pole, given that 1 cm^3 of iron has approximately 8 g mass. (Use π = 3.14)

✔ ANSWER

Lower cylinder: r = 12 cm, h = 220 cm. Volume = 3.14 × 12 × 12 × 220 = 99475.2 cm^3

Upper cylinder: r = 8 cm, h = 60 cm. Volume = 3.14 × 8 × 8 × 60 = 12057.6 cm^3

Total volume = 99475.2 + 12057.6 = 111532.8 cm^3

Mass = 111532.8 × 8 g = 892262.4 g

Mass of the pole ≈ 892.26 kg

7 **A solid consisting of a right circular cone of height 120 cm and radius 60 cm standing on a hemisphere of radius 60 cm is placed upright in a right circular cylinder full of water such that it touches the bottom. Find the volume of water left in the cylinder, if the radius of the cylinder is 60 cm and its height is 180 cm.**

✔ ANSWER

Volume of cylinder = $\pi r^2 h$ = π × 60 × 60 × 180 = 648000π cm^3

Volume of cone = $\frac{1}{3}\pi r^2 h$ = ⅓ × π × 60 × 60 × 120 = 144000π cm^3

Volume of hemisphere = $\frac{2}{3}\pi r^3$ = ⅔ × π × 60^3 = 144000π cm^3

Volume of solid = $144000\pi + 144000\pi = 288000\pi$ cm³

Water left = $648000\pi - 288000\pi = 360000\pi = 360000 \times 22/7 \approx 1131428.57$ cm³

Volume of water left ≈ 1131428.57 cm³ ≈ 1.131 m³

8 **A spherical glass vessel has a cylindrical neck 8 cm long, 2 cm in diameter; the diameter of the spherical part is 8.5 cm. By measuring the amount of water it holds, a child finds its volume to be 345 cm³. Check whether she is correct, taking the above as the inside measurements, and π = 3.14.**

✔ ANSWER

Neck (cylinder): r = 1 cm, h = 8 cm. Volume = $3.14 \times 1 \times 1 \times 8 = 25.12$ cm³

Sphere: r = 8.5/2 = 4.25 cm. Volume = $4/3 \times 3.14 \times (4.25)^3 = 4/3 \times 3.14 \times 76.765625 \approx 321.39$ cm³

Total volume = $25.12 + 321.39 = 346.51$ cm³

She is not correct. The correct volume is 346.51 cm³ (not 345 cm³).

🎉 **Chapter 12 done! Revise the answers once more tomorrow.**